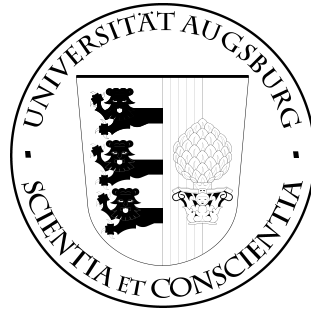


INSTITUTE OF COMPUTER SCIENCE UNIVERSITY OF AUGSBURG



Bachelor's Thesis

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Abstract

Describe the content of the present thesis here. Briefly explain the starting point of the thesis and outline its objectives. Then, discuss the methods used and the results achieved.

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1. Introduction

2. Theoretical Fundamentals of Quantum Computing

2.1. Basics

2.1.1. Quantum States

In an n -dimensional Hilbert space with a chosen orthonormal basis, a quantum state, denoted by the ket $|\psi\rangle$, can be represented as an n -dimensional complex column vector $\vec{\psi}$:

$$|\psi\rangle \doteq \vec{\psi} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}. \quad (2.1)$$

Its corresponding dual state, denoted by the bra $\langle\psi|$, is represented by the conjugate transpose of this column vector, resulting in an n -dimensional row vector $\vec{\psi}^\dagger$:

$$\langle\psi| \doteq \vec{\psi}^\dagger = (c_1^* \quad c_2^* \quad \dots \quad c_n^*) \quad (2.2)$$

[1, p.16f.], [3, p. 10f.].

To ensure consistency with the probability interpretation, quantum states must be normalized to unity. Therefore, for a quantum state $|\psi\rangle$, the sum of the absolute squares of its probability amplitudes must equal one:

$$|c_1|^2 + |c_2|^2 + \dots + |c_n|^2 = 1 \quad (2.3)$$

[1, p.5].

2.1.2. Operators and Observables

A linear operator acts on a ket from the left to produce another ket vector, or on a bra from the right to produce another bra vector:

$$X |\alpha\rangle = |\beta\rangle, \quad (2.4)$$

$$\langle\alpha| X = \langle\beta|. \quad (2.5)$$

An operator X is defined as the **null operator** if, for any arbitrary ket $|\alpha\rangle$, it satisfies:

$$X |\alpha\rangle = 0. \quad (2.6)$$

2. Theoretical Fundamentals of Quantum Computing

Operators can be added ($X + Y = Y + X$) and multiplied. While operator multiplication is generally noncommutative ($XY \neq YX$), it is strictly associative:

$$X(YZ) = (XY)Z = XYZ \quad (2.7)$$

[3, p.11, 13-15].

Because quantum states are represented as vectors in $\mathbb{C}^n$, a linear operator can be represented by a complex $n \times n$ matrix U . Given an input ket $|\psi\rangle$, the resulting state $|\phi\rangle$ is calculated via matrix-vector multiplication:

$$|\phi\rangle = U |\psi\rangle. \quad (2.8)$$

Crucially, physical operations that evolve a quantum state must preserve its probability normalization. Consequently, physical evolution operators are restricted to unitary matrices, ensuring that the inner product—and thus the normalization—remains invariant [1, p.19].

An **observable** represents a physical quantity of a system that can be experimentally measured. Classical examples include position, momentum, angular momentum, and energy. In an N -qubit system, an observable is defined mathematically as a $2^N \times 2^N$ Hermitian matrix H . Because the operator is Hermitian, it guarantees that all possible measurement outcomes are strictly real numbers and correspond directly to the eigenvalues of H [3, p.16f.], [4, p.8]. Crucially, the act of measurement fundamentally alters the system: upon obtaining a specific eigenvalue, the quantum state instantaneously collapses into the corresponding eigenstate of that observable [3, p.22].

Due to the inherent probabilistic nature of quantum mechanics, we generally cannot predict the outcome of a single measurement with absolute certainty. Instead, we evaluate the **expectation value**, which represents the statistical average of measured values over a large ensemble of identically prepared systems [3, p. 23].

In the context of this thesis, the primary observables of interest are Hamiltonians, which represent the total energy of the physical system. Let H be the Hamiltonian. We conventionally assume that its resulting energy eigenvalues are ordered from the ground state upwards ($E_0 \leq E_1 \leq E_2 \leq \dots$), satisfying the eigenvalue equation:

$$H |E_i\rangle = E_i |E_i\rangle. \quad (2.9)$$

Given a hamiltonian Operator H and a quantum state $|\psi\rangle$ the expectation value can be calculated as:

$$\langle H \rangle_\psi := \langle \psi | H | \psi \rangle \quad (2.10)$$

[1, p.42]

2.1.3. Unitary Operations and Generators

The physical transformations applied to quantum states are represented by unitary operators. To construct parameterized quantum gates, these unitary operations are typically expressed via the exponential map:

$$U(\theta) = e^{-i\frac{\theta}{2}G}, \quad (2.11)$$

where θ is a real-valued scalar parameter (such as a time or rotation angle), and G is a Hermitian matrix ($G = G^\dagger$) known as the *generator* of the operation. [1, p.34, 39] [2, p.1]

2.1.4. Tensor Products

2.2. Fermionics

2.2.1.

3. Excitations

General overview

3.1. Qubit Excitations

Definition and Explanation

3.2. Fermionic Excitations

Definition and Explanation

3.3. Fermionic systems

- Fermions, their challenges and their differences to qubits
- Fermionic annihilation and creation operator: $a_i^\dagger$ und a_i .

4. Different design approaches

4.1. Approach A: Operator-Algebra

- Concept: Implementation of the exact Algebra
- Structure: Addition, subtraction und multiplication of chained operators.

This approach tries to follow the exact definition of the algebra operators. It begins with defining the operators itself. A function is defined that encodes the logic of the operators. Given a specific state it calculates how the operator by definition acts on this state. Given the operator $a_j^\dagger$ and the state ϕ it would act as following.

$$a_j^\dagger \phi = a_j^\dagger |n_0, n_1, \dots, n_j, \dots\rangle = (1 - n_j)(-1)^{\sum_{k=0}^{j-1} n_k} |n_0, n_1, \dots, 1_j, \dots\rangle \quad (4.1)$$

4.2. Approach B: Generator-based evolution (symbolically)

- Concept: Apply Generator directly on a state
- Unitary operation: How to calculate $U = \exp\left(\sum \theta_{pq} \hat{E}_{pq}\right)$ efficient?

5. Implementation in C++

- Explanation
- Decision on implementation
- Software architecture
- Effizienz strategy:
 - Why C++ (Memory management, Speed, etc.)
 - Data structures
 - Optimization of the loop

6. Evaluation and results

- Check for Correctness with Tequila
- Performance-analyse: how does the computation scale with difference states
- Comparison of different approaches

7. Summary and Outlook

List of Figures

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A. Appendix Chapter

If you want to add information to the thesis that is only marginally relevant for understanding your work, create an appendix for this purpose. Internet reference [?], reference with multiple authors [?]

A.1. foo

A.2. bar

B. Second Appendix Chapter

